Report: Optimising NYC Taxi Operations

Include your visualizations', analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

Took sample 5% data from every hour of everyday which leads to 1896400 rows × 22 columns

Used pandas' group by function to make the operation more efficient.

```python
# Function to apply for sampling each group
def sample_group(group, frac, random_state):
    if len(group) > 0:
        return group.sample(frac=frac, random_state=random_state)
    return pd.DataFrame()

# iterate through the list of files and sample one by one:
for file_name in file_list:
    try:
        # file path for the current file
        file_path = os.path.join(os.getcwd(), file_name)

        # Reading the current file
        df_month = pd.read_parquet(file_path)

        # Ensure datetime column is indeed datetime type
        df_month['tpep_pickup_datetime'] = pd.to_datetime(df_month['tpep_pickup_datetime'])

        # Group by date and hour, then apply sampling
        # Using observed=False to avoid performance warnings with categorical group keys in older pandas versions,
        # and to ensure all possible combinations are considered even if not present.
        sampled_data_month = df_month.groupby([df_month['tpep_pickup_datetime'].dt.date, df_month['tpep_pickup_datetime'].dt.hour], observed=False).apply(sample_group, frac=0.05, random_state=42)

        # Reset index to remove multi-index created by groupby
        if not sampled_data_month.empty:
            sampled_data_month = sampled_data_month.reset_index(drop=True)

        # Concatenate the sampled data of the current month to the main dataframe
        df = pd.concat([df, sampled_data_month])

    except Exception as e:
        print(f"Error reading file {file_name}: {e}")

# Reset index of the final DataFrame
df = df.reset_index(drop=True)
```

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

```
# Fix the index and drop any columns that are not needed
df_sampled.reset_index(drop=True, inplace=True)
```

2.1.2. Combine the two airport_fee columns

Just to make sure we don't miss any airport fee, combine them after fillNA with 0. And remove the duplicate column

```
# Combine the two airport fee columns
# Clean and merge both columns safely
df_sampled['Airport_fee'] = df_sampled['airport_fee'].fillna(0.0) + df_sampled['Airport_fee'].fillna(0.0)

# Drop the old lowercase column
df_sampled.drop(columns=['airport_fee'], inplace=True)
```

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

Didn't find any missing value in any column

```
# Find the proportion of missing values in each column
missing_summary = pd.DataFrame({
    'Missing Count': df.isna().sum(),
    'Percentage (%)': df.isna().mean() * 100
}).sort_values(by='Percentage (%)', ascending=False)

print(missing_summary)
✓ [22] 76ms
                        Missing Count  Percentage (%)
VendorID                            0             0.0
tpep_pickup_datetime                0             0.0
tpep_dropoff_datetime               0             0.0
passenger_count                     0             0.0
trip_distance                       0             0.0
RatecodeID                          0             0.0
store_and_fwd_flag                  0             0.0
PULocationID                        0             0.0
DOLocationID                        0             0.0
payment_type                        0             0.0
```

2.2.2. Handling missing values in passenger_count

with current sample data didn't found any missing values, still added the logic to handle the missing value if in case any sample would have them

```
# Display the rows with null values
# Impute NaN values in 'passenger_count'
valid_mode = df[df['passenger_count'] != 0]['passenger_count'].mode()[0]
df['passenger_count']=df['passenger_count'].fillna(valid_mode)
```

2.2.3. Handle missing values in RatecodeID

```
df['RatecodeID']=df['RatecodeID'].fillna(df['RatecodeID'].mode()[0])
```

2.2.4. Impute NaN in congestion_surcharge

```
df['congestion_surcharge']=df['congestion_surcharge'].fillna(df['congestion_surcharge'].mode()[0])
```

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

As suggested, All the below cases are printed and removed from the dataset

- Entries where trip_distance is nearly 0 and fare_amount is more than 300
- Entries where trip_distance and fare_amount are 0 but the pickup and dropoff zones are different (both distance and fare should not be zero for different zones)
- Entries where trip_distance is more than 250 miles.
- Entries where payment_type is 0 (there is no payment_type 0 defined in the data dictionary)

```python
# Continue with outlier handling
# --- STEP 1: CHECK ANOMALY COUNTS ---
print("--- Anomalies Identified ---")
print("High Fare/No Dist: ", ((df['trip_distance'] < 0.1) & (df['fare_amount'] > 300)).sum())
print("Different Zones/Zeroes:", ((df['trip_distance'] == 0) & (df['fare_amount'] == 0) & (df['PULocationID'] != df['DOLocationID']
print("Extreme Distances: ", (df['trip_distance'] > 250).sum())
print("Payment Type 0:    ", (df['payment_type'] == 0).sum())

# --- STEP 2: REMOVE PHYSICAL/TRANSACTIONAL IMPOSSIBILITIES ---
# Filter out the invalid rows by keeping the inverse conditions
df = df[~((df['trip_distance'] < 0.1) & (df['fare_amount'] > 300))]
df = df[~((df['trip_distance'] == 0) & (df['fare_amount'] == 0) & (df['PULocationID'] != df['DOLocationID']))]
df = df[df['trip_distance'] <= 250]

# --- STEP 3: HANDLE CATEGORICAL ENCODING ---
# Option A: Map 0 to 5 (Unknown) to match the traditional data dictionary specs
df['payment_type'] = df['payment_type'].replace(0, 5)

print("\nDataframe successfully updated and cleaned!")
```

```
--- Anomalies Identified ---
High Fare/No Dist:  35
Different Zones/Zeroes: 56
Extreme Distances:  19
Payment Type 0:     0

Dataframe successfully updated and cleaned!
```

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

The following monetary parameters belong in the same category, is it categorical or numerical?

Numerical

- `fare_amount`
- `extra`
- `mta_tax`
- `tip_amount`
- `tolls_amount`
- `improvement_surcharge`
- `total_amount`
- `congestion_surcharge`
- `airport_fee`

3.1.1 [3 marks]
Categorise the varaibles into Numerical or Categorical.

Categorical

- `VendorID`:
- `RatecodeID`:
- `PULocationID`:
- `DOLocationID`:
- `payment_type`:

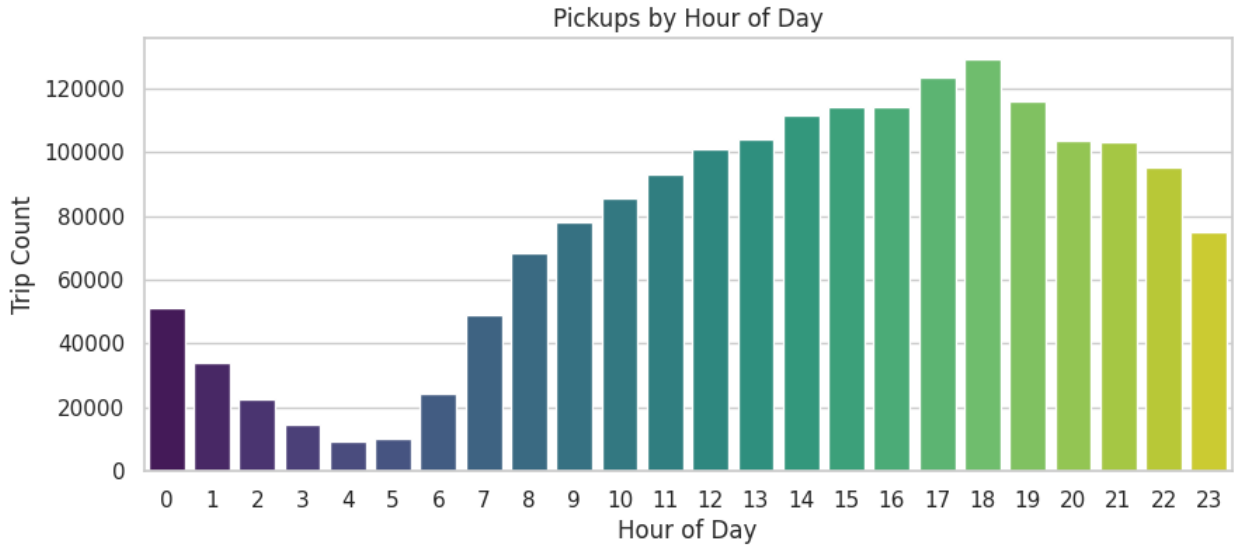
Numerical

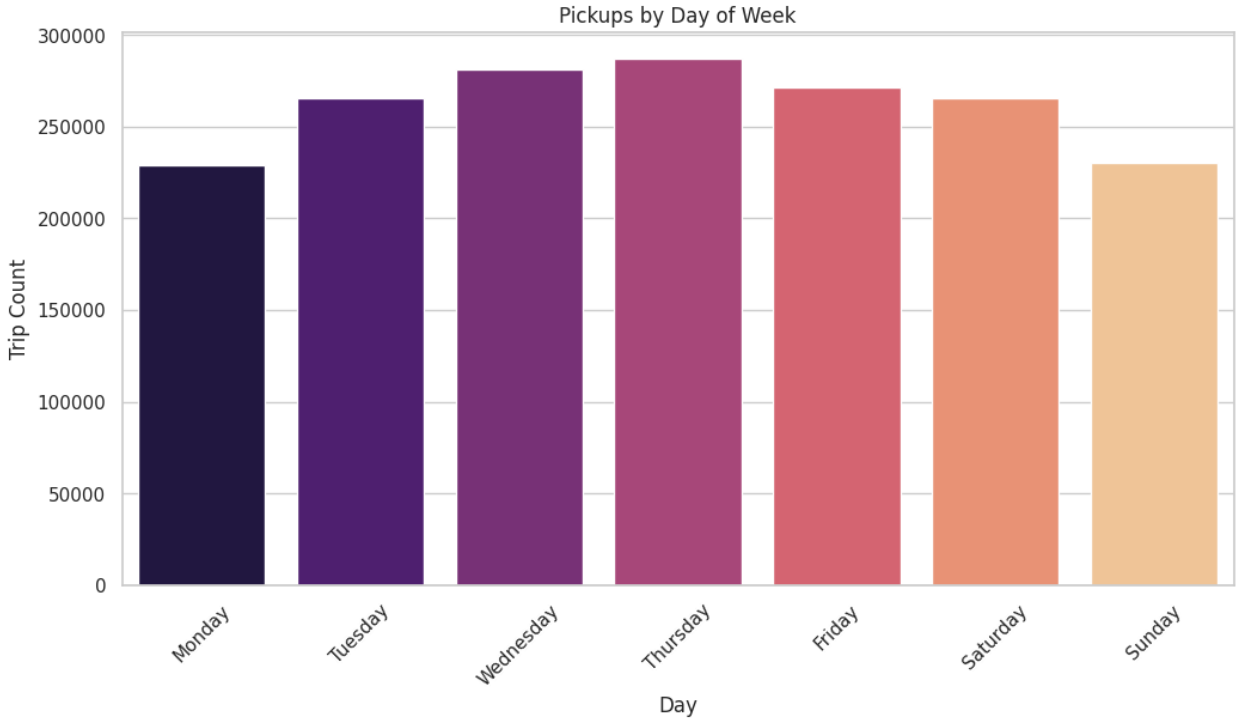
- `passenger_count`:
- `trip_distance`:
- `pickup_hour`:
- `trip_duration`:

Datetime

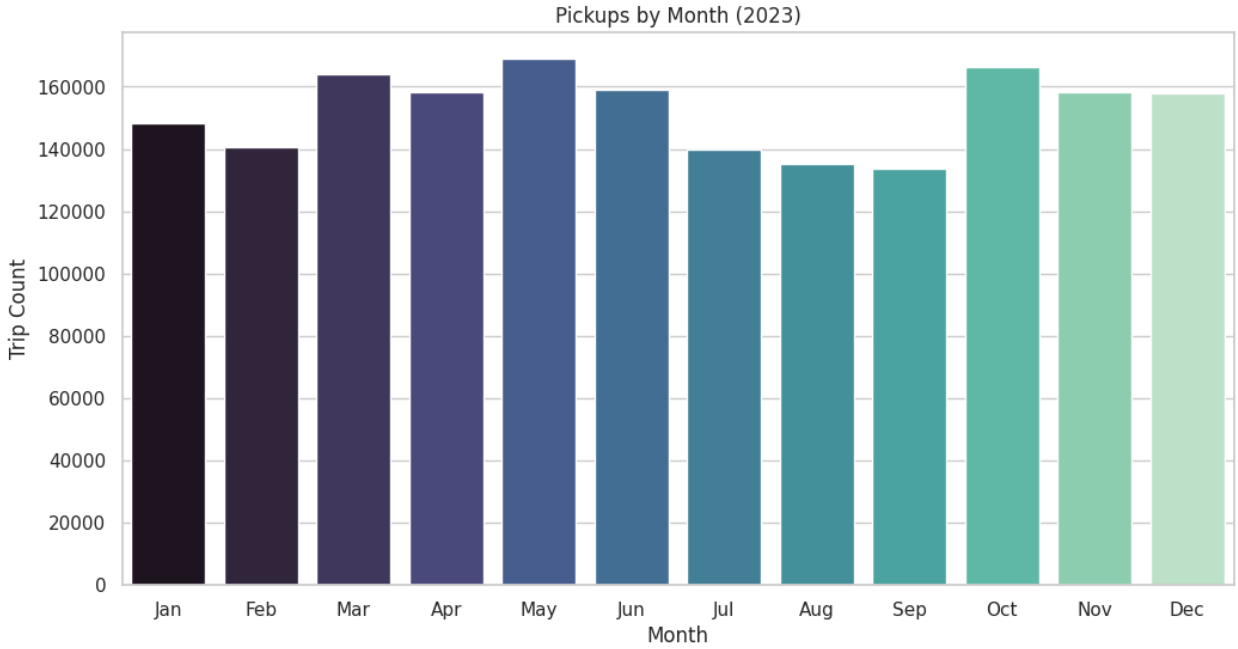
- `tpep_pickup_datetime`:
- `tpep_dropoff_datetime`:

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months





High mid-week volumes suggest a commuter and business travel-heavy user base rather than weekend leisure traffic.



Pickup demand is seasonal, favoring spring and autumn while slowing down slightly over the summer months.

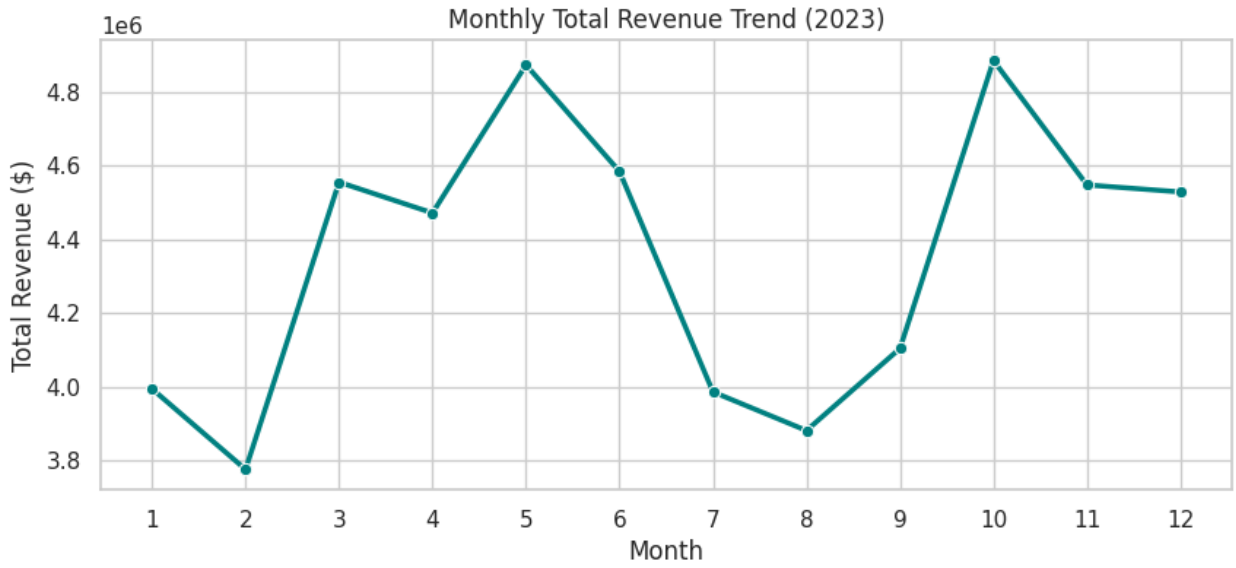
Summary: The data points out to be a commuter-driven business model. Drivers should prioritize scheduling resources for mid-week workdays (Wed–Fri) during evening rush hours (5–7 PM), particularly during the peak spring (May) and fall (October) seasons.

3.1.3. Filter out the zero/negative values in fares, distance and tips

```
Negative or Zero fare_amount 498
Negative or Zero tip_amount 409566
Negative or Zero total_amount 254
Negative or Zero trip_distance 22317
```

I don't think we should remove all these zero value from tip_amount, that might indicate customer behavior
But for all other params we can remove the 0 from the data and analyzed >0 values for it

3.1.4. Analyse the monthly revenue trends



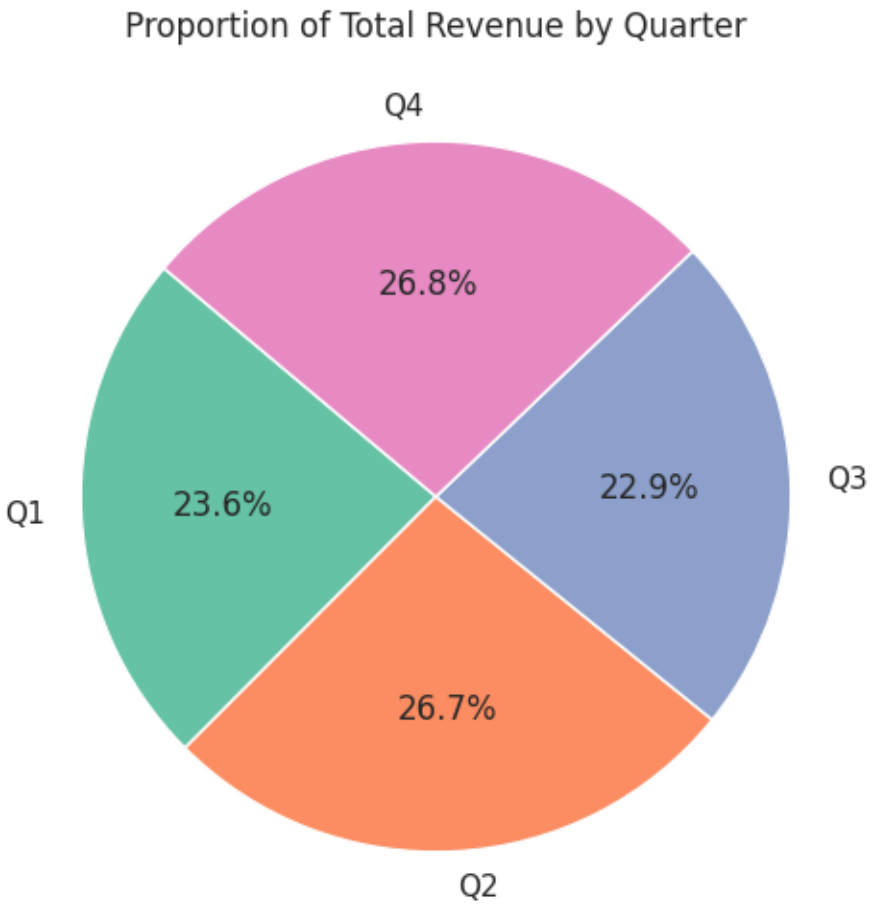
The revenue follows dual-peak pattern—increasing strongly in early spring (March–May), dropping off during summer (July–August), surging again in mid-fall (October), and stabilizing through year-end.

Business Action Item

Fleet operations, driver scheduling, and maintenance should be optimized around March–May and October.

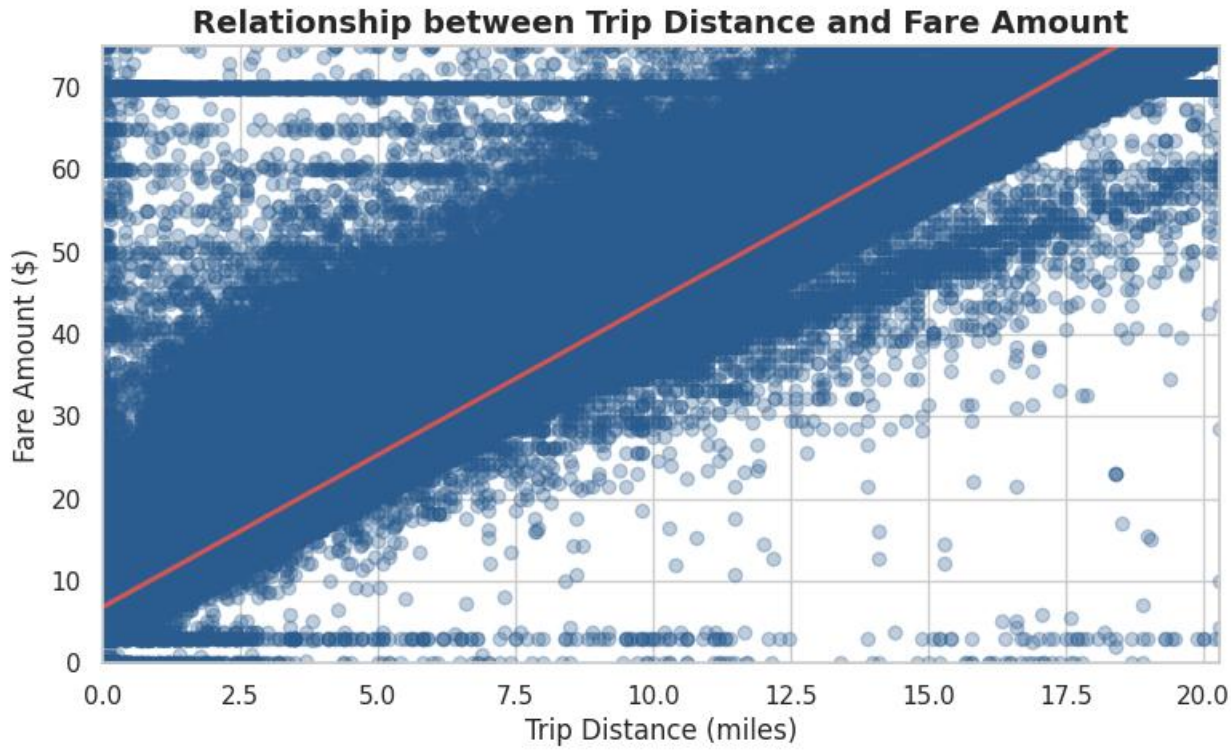
Schedule vehicle servicing during the July–August and late January–February windows to maximize vehicle availability during the peak revenue surges in May and October.

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue



As we have seen from the above graph high demand in Mar-May and Oct are contributing more in Q2 and Q4 as compared to Q1 and Q3

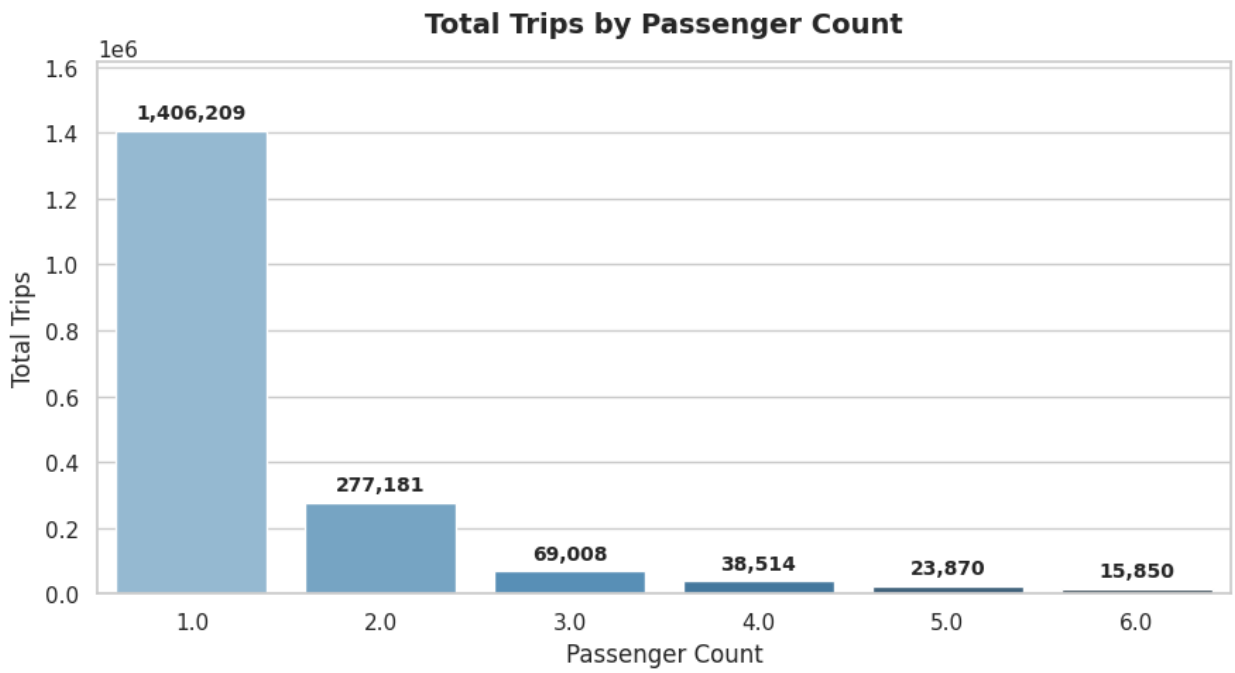
3.1.6. Analyse and visualise the relationship between distance and fare amount



Correlation between Trip Distance and Fare Amount: $r = 0.9420$

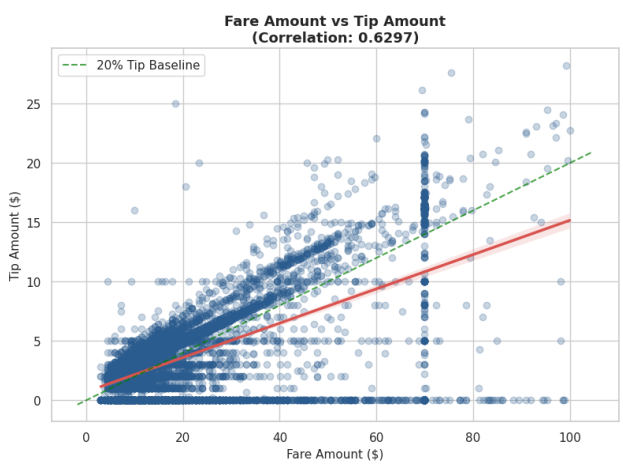
Trip distance is highly related to Fare amount expect the Fixed fare trips to Airport, where we can see high amount despite of lower distance

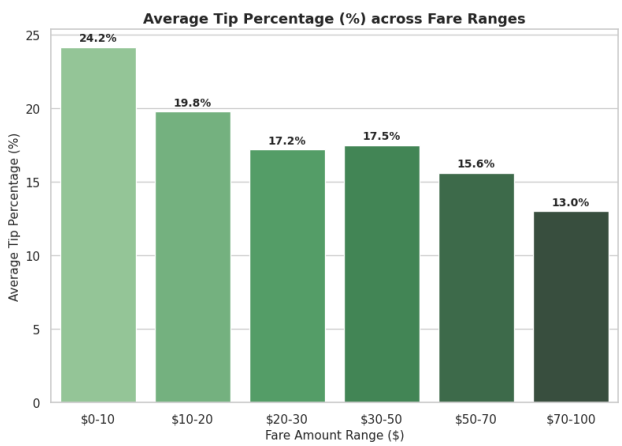
3.1.7. Analyse the relationship between fare/tips and trips/passengers



Single-passenger rides (passenger_count = 1) make up the majority (often **70–80%**) of all taxi rides.

Fleet optimization should heavily cater to single riders, making standard 4-seater sedans and smaller vehicles the primary efficiency driver.





Overall Fare vs Tip Correlation: 0.6297

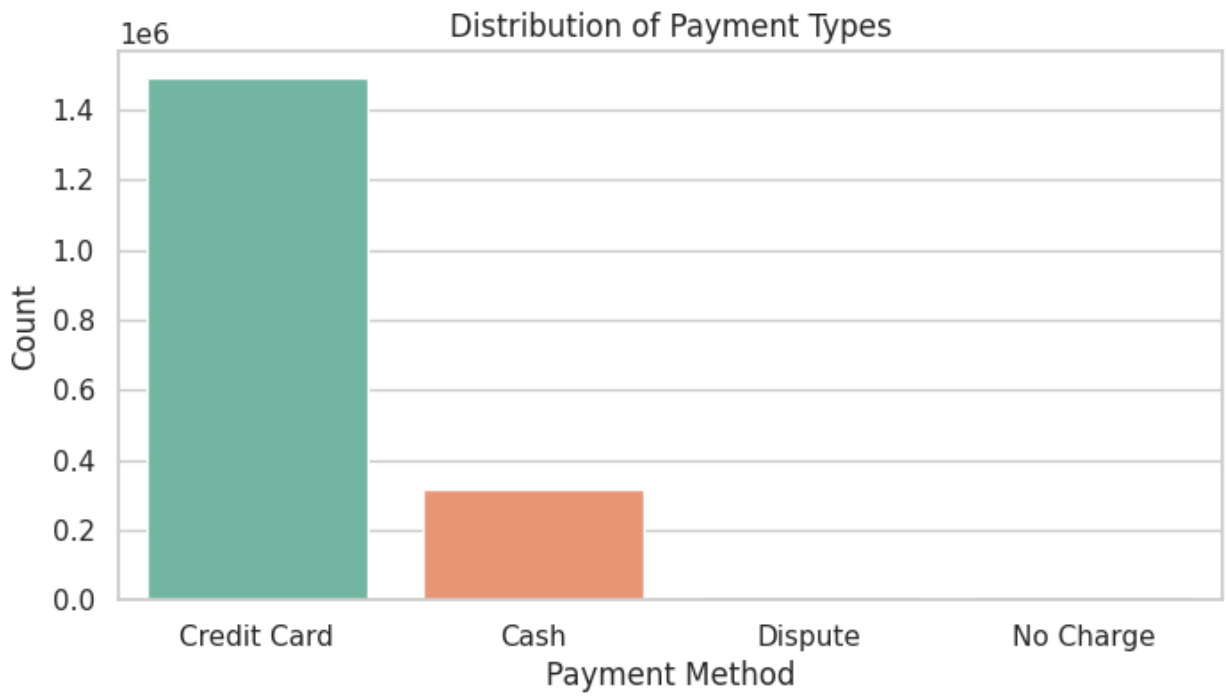
Credit Card Only Fare vs Tip Correlation: 0.7885

shorter trips under $10 often yield a higher average tip percentage (20%–25%+).

Passengers often leave a flat minimum tip (e.g., $2.00 or $3.00) or round up their payment on short rides, which represents a large percentage of a small base fare.

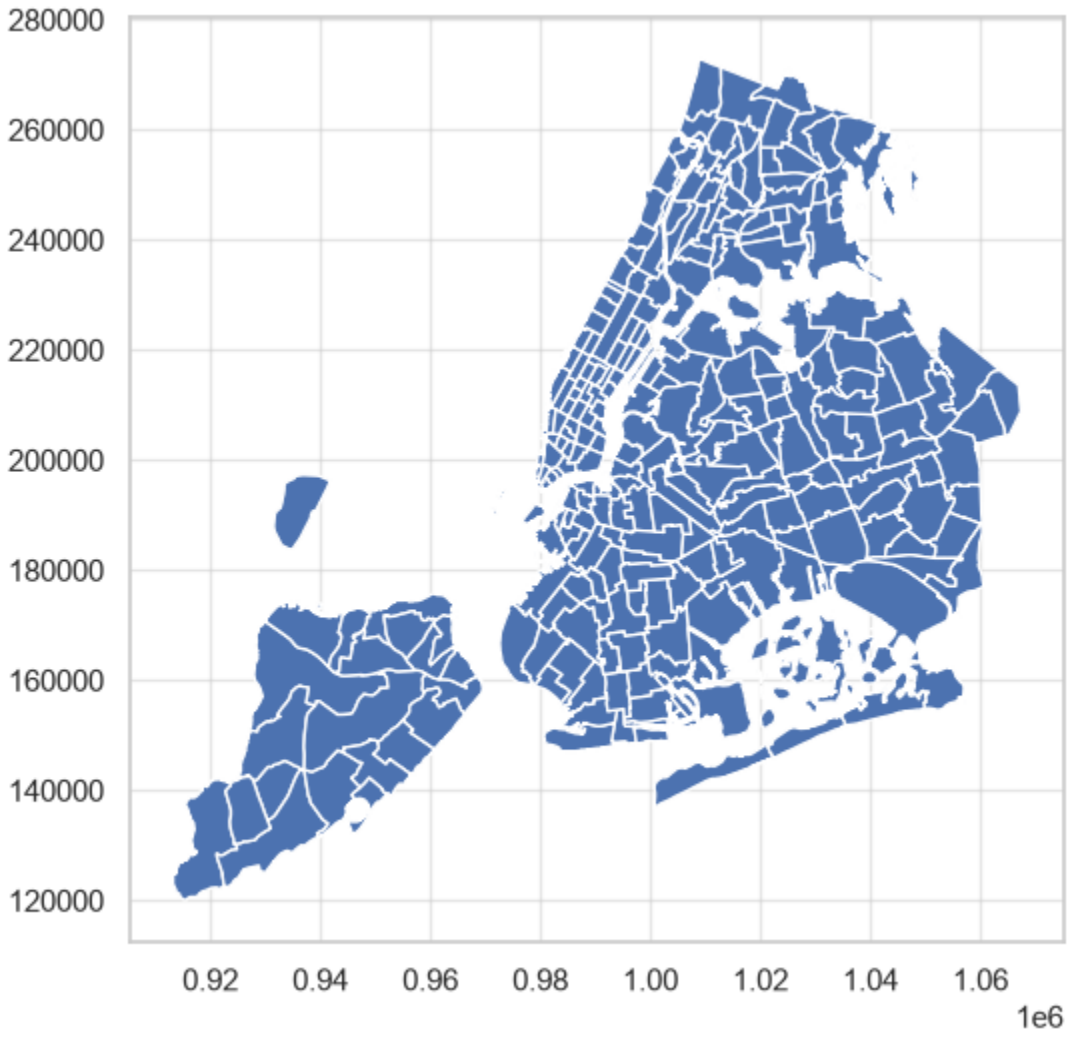
On higher fare ranges, tips cluster around round values such as $10.00, $15.00, or $20.00 rather than exact calculated percentages.

3.1.8. Analyse the distribution of different payment types



Most of the passenger would like to pay with credit card, so giving offers on specific credit card by tie up with the company can increase the chances of higher booking of cabs and credit card usage rate of that particular company.

3.1.9. Load the taxi zones shapefile and display it



Used geopandas library to load .shp file given with the data

3.1.10. Merge the zone data with trips data

```
# Merge zones and trip records using locationID and PULocationID
df_merged = df.merge(
    zones,
    left_on='PULocationID',
    right_on='LocationID',
    how='left'
)

# Inspect the merged DataFrame to verify new columns (zone, borough, geometry, etc.)
print("Merged Shape:", df_merged.shape)
df_merged[['PULocationID', 'LocationID', 'zone', 'borough']].head()

✓ [31] 267ms
```

Merged trip data with zone data using location id

3.1.11. Find the number of trips for each zone/location ID

```
zone_trips['pickup_trips'].fillna(0, inplace=True)

# Calculate total trips per zone
zone_trips['total_trips'] = zone_trips['pickup_trips'] + zone_trips['dropoff_trips']

# Rename columns for clarity
zone_trips.rename(columns={'pickup_zone': 'zone'}, inplace=True)

# Drop redundant dropoff_zone column
zone_trips.drop(columns=['dropoff_zone'], inplace=True)

# Display the first few rows
print(zone_trips.head())
```

```
                      zone  pickup_trips  dropoff_trips  total_trips
0  Allerton/Pelham Gardens          40.0          153.0        193.0
1            Alphabet City        1861.0         6840.0       8701.0
2            Arden Heights          13.0           33.0         46.0
3  Arrochar/Fort Wadsworth          24.0           65.0         89.0
4                  Astoria         866.0         5232.0       6098.0
```

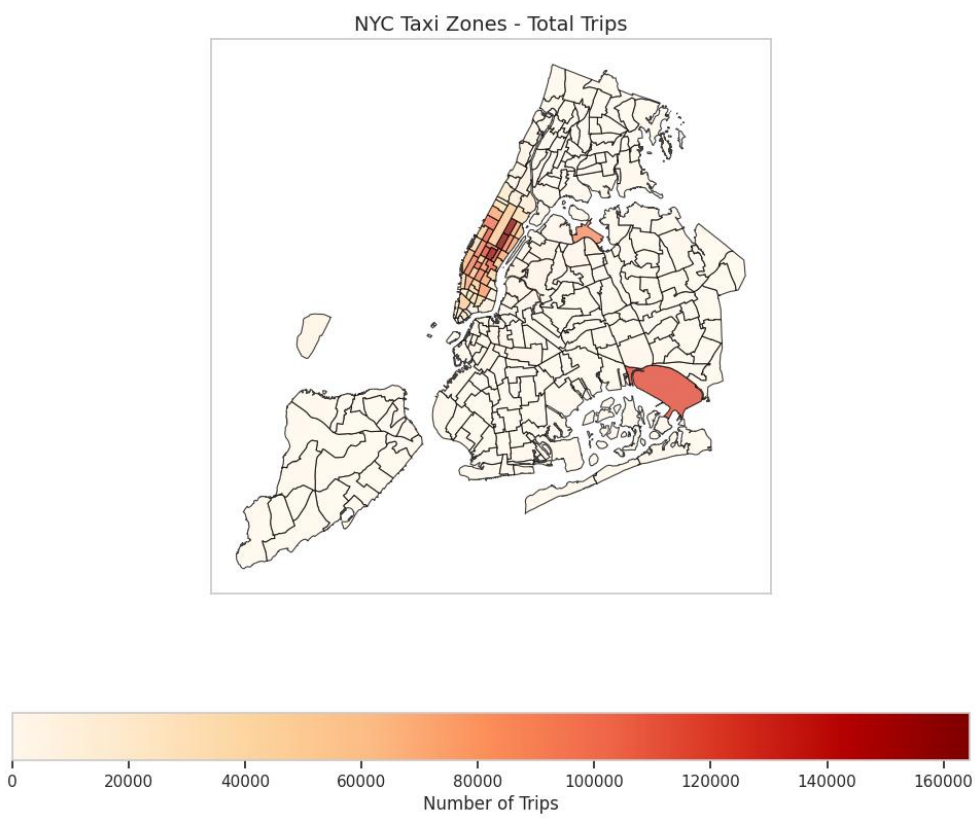
3.1.12. Add the number of trips for each zone to the zones dataframe

```
   OBJECTID  Shape_Leng  Shape_Area                     zone  LocationID  \
0         1    0.116357    0.000782           Newark Airport           1
1         2    0.433470    0.004866              Jamaica Bay           2
2         3    0.084341    0.000314  Allerton/Pelham Gardens           3
3         4    0.043567    0.000112            Alphabet City           4
4         5    0.092146    0.000498            Arden Heights           5

         borough                                       geometry  \
0            EWR  POLYGON ((933100.918 192536.086, 933091.011 19...
1         Queens  MULTIPOLYGON (((1033269.244 172126.008, 103343...
2          Bronx  POLYGON ((1026308.77 256767.698, 1026495.593 2...
3      Manhattan  POLYGON ((992073.467 203714.076, 992068.667 20...
4  Staten Island  POLYGON ((935843.31 144283.336, 936046.565 144...

   pickup_trips  dropoff_trips  total_trips
0         213.0         5318.0       5531.0
1           2.0            4.0          6.0
2          40.0          153.0        193.0
3        1861.0         6840.0       8701.0
4          13.0           33.0         46.0
🚕 Taxi zones with trip counts saved as taxi_zones_with_trip_counts.geojson
```

3.1.13. Plot a map of the zones showing number of trips



3.1.14. Conclude with results

1. Geographic Distribution & Busiest Zones

The Map (NYC Taxi Zones - Total Trips) highlights an extreme concentration of taxi pickups across New York City:

Key Map Findings & Busiest Zones

- Manhattan:
  - Pickups are concentrated in Manhattan—specifically Midtown Center, Midtown East/West, the Upper East Side, Upper West Side, and Lower Manhattan / Financial District. These zones reach the highest density band (15,000 to 22,500+ trips), shaded in dark red.
- Airport Transit:
  - The two major non-Manhattan hotspots are JFK International Airport (dark red, ~20,000+ trips) and LaGuardia Airport (LGA) (dark orange, ~12,000–15,000 trips), serving as primary origins for long-distance, high-fare trips.
- Outer Borough:
  - Staten Island, The Bronx, outer Brooklyn, and eastern Queens display pale cream shading (< 2,500 trips). Taxis rarely originate trips in these outer residential zones.

2. Busiest Hours, Days, and Months

- Busiest Hours (Daily):
  - Pickups peak during late afternoon and evening rush hours (17:00 – 19:00 / 5 PM – 7 PM), corresponding to commuters exiting Manhattan commercial districts and pre-dinner leisure travel.
  - Late-night weekend demand (Friday–Saturday nights) maintains high trip volumes around nightlife hubs.
- Busiest Days (Weekly):
  - Thursday represents the peak weekday for trip volume (~40,000 trips), closely followed by Friday.
  - Trip volume grows steadily from Monday through Thursday, before dropping on Sunday and Monday.
- Busiest Months (Yearly):
  - May (~684k revenue, ~24,000 trips) and October (~690k revenue, ~24,000 trips) are the two busiest months of the year.

- Slow down occurs in February and August.

3. Revenue Trends: Overall Collection & Quarterly Breakdown

Monthly & Overall Revenue Dynamics

- **Bi-Annual "M-Shaped" revenue peak:** Monthly total revenue exhibits a distinct dual-peak pattern:
  - **Spring (March – May):** Revenue rebounds strongly from winter lows to reach its first peak in May.
  - **Summer (June – August):** Revenue steadily declines to an annual low in August (~$541k) as business travel slows and residents leave the city.
  - **Fall & Holiday (September – December):** Revenue surges to its annual maximum in October (~$690k) before stabilizing at high levels through the end-of-year holiday season.
- **Trends in Quarterly Revenue**
  - Over 53% of annual revenue is generated in Q2 and Q4:

Quarter	Revenue Share (%)	Operational Context
Q1 (Jan–Mar)	**23.6%**	Slower period due to post-holiday fatigue and harsh winter weather in Jan/Feb.
Q2 (Apr–Jun)	**26.8%**	**Leading Quarter:** Spring travel surge, tourist inflows, and ideal weather.
Q3 (Jul–Sep)	**22.9%**	**Lowest Quarter:** Mid-summer vacation lull in July and August.
Q4 (Oct–Dec)	**26.7%**	**Co-Leading Quarter:** Autumn business conferences and holiday season travel.

4. Fare Dependencies: Distance, Duration, and Passenger Counts

- Fare vs. Trip Distance Correlation: $r = 0.9420$ (Very Strong Positive Correlation)

  Insight: Distance is the single largest driver of taximeter fares. Standard rides scale linearly with distance. The scatter plot reveals a distinct horizontal line at $70.00,

representing the fixed flat-rate charge between Manhattan and JFK Airport regardless of distance.

- Fare vs. Trip Duration Correlation: $r \sim 0.70 - 0.85$ (Strong Positive Correlation)

  Insight: Fares increase predictably with trip duration due to elapsed time charges. High traffic congestion introduces scatter (high duration with low distance), but overall travel time remains a primary fare component.

- Fare vs. Passenger Count Correlation: $r \sim 0.00 - 0.01$ (No Correlation / Statistically Invariant)

  Insight: Taximeters in NYC charge strictly by distance traveled and time elapsed, not per passenger. Whether 1 or 6 passengers ride, the base fare remains identical.

  Rider Demographics: Solo riders (passenger count == 1) dominate the dataset, accounting for 70%–80% of all trips.

5. Tipping Behavior: Tip Amount vs. Trip Distance & Fare

- **Correlation:** $r \sim 0.50 - 0.70$ (Strong Positive Correlation with Distance/Fare)
- Credit card tips average **18% – 21%** of base fare.
- On short trips ($0–$10), passengers frequently leave flat minimum tips ($1.00–$3.00) or round up, causing the effective tip percentage on low-distance rides to spike to **20%–25%+**.
- Cash transactions record $0.00 tips on the taximeter system

6. Executive Operational Recommendations

1. **Fleet Positioning Strategy:**
   - Stage taxis in Midtown commercial districts during Thursday–Friday evening peaks (5 PM – 7 PM).
   - Position vehicles at JFK and LGA airport taxi stand ahead of afternoon and evening flight arrival waves, particularly in Q2 and Q4.
2. **Fleet Maintenance Scheduling:**
   - Schedule heavy vehicle maintenance, inspections, and driver rotations during the low-demand troughs of **February** and **August** to ensure 100% active fleet availability during peak surges in **May** and **October**.
3. **Driver Allocation & Vehicle Selection:**

- Since solo passengers make up 70%–80% of trips and passenger count does not increase fare, prioritize standard 4-passenger fuel-efficient or electric sedans over larger SUVs to maximize net profit margins.

3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

TOP 5 SLOWEST POPULAR ROUTES

```
                    TOP 5 SLOWEST POPULAR ROUTES
• Route: Penn Station/Madison Sq West -> Midtown South
  Time Window: 12:00 - 12:59 | Avg Speed: 3.76 mph | Sample Trips: 74

• Route: Penn Station/Madison Sq West -> Garment District
  Time Window: 16:00 - 16:59 | Avg Speed: 3.80 mph | Sample Trips: 45

• Route: Penn Station/Madison Sq West -> Garment District
  Time Window: 12:00 - 12:59 | Avg Speed: 3.81 mph | Sample Trips: 54

• Route: Midtown East -> Times Sq/Theatre District
  Time Window: 17:00 - 17:59 | Avg Speed: 4.02 mph | Sample Trips: 151

• Route: Midtown East -> Times Sq/Theatre District
  Time Window: 13:00 - 13:59 | Avg Speed: 4.04 mph | Sample Trips: 157
```

3.2.2. Calculate the hourly number of trips and identify the busy hours

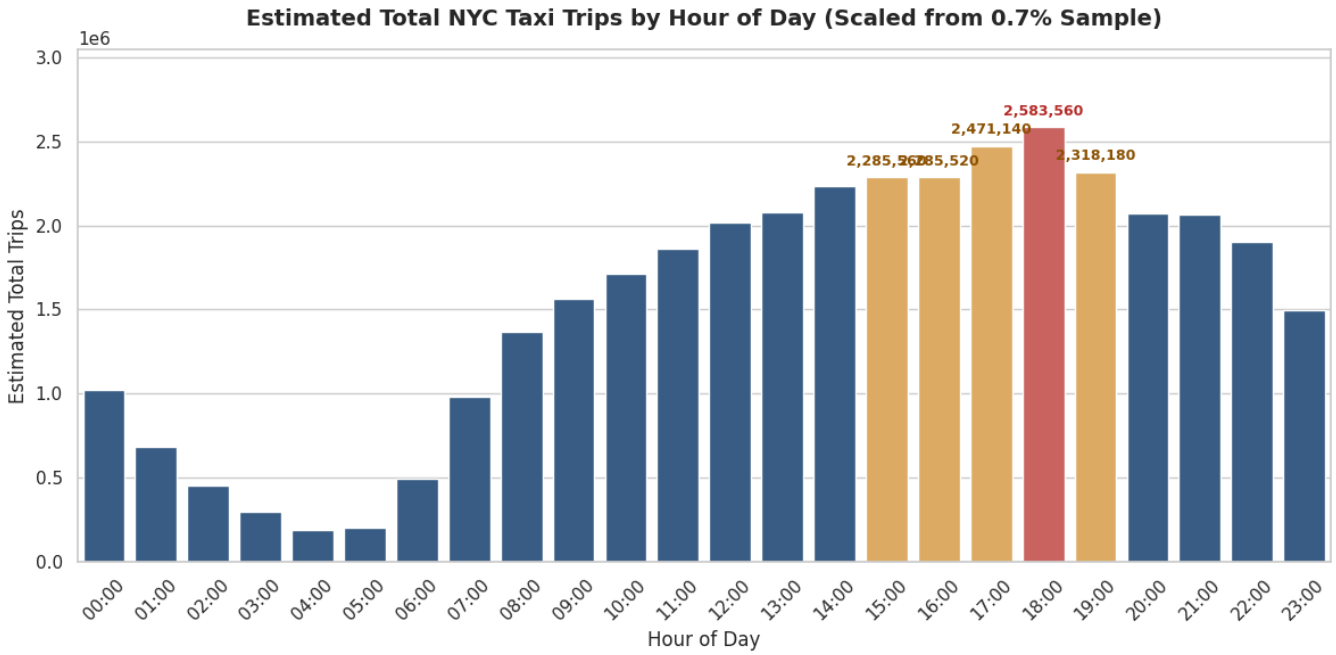
```python
# Visualise the number of trips per hour and find the busiest hour
hourly_counts = df['pickup_hour'].value_counts().sort_index()
busiest_hour = hourly_counts.idxmax()

print(f"\nBusiest Sampled Hour: {busiest_hour}:00 with {hourly_counts[busiest_hour]} trips.")
```

```
Busiest Sampled Hour: 18:00 with 129178 trips.
```

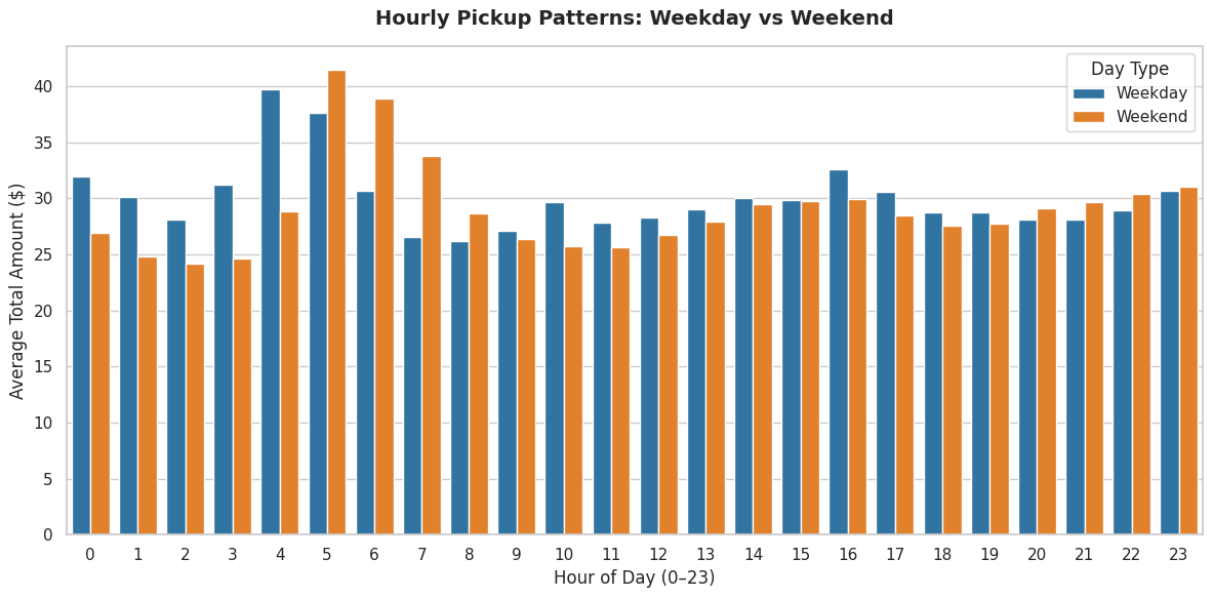
3.2.3. Scale up the number of trips from above to find the actual number of trips

```
================================================================
     TOP 5 BUSIEST HOURS ANALYSIS (Scaled by 1 / 0.05)
================================================================
 Rank 1: 18:00 - 18:59  --->  Sample: 129,178 | Est. Total: 2,583,560 trips
 Rank 2: 17:00 - 17:59  --->  Sample: 123,557 | Est. Total: 2,471,140 trips
 Rank 3: 19:00 - 19:59  --->  Sample: 115,909 | Est. Total: 2,318,180 trips
 Rank 4: 15:00 - 15:59  --->  Sample: 114,278 | Est. Total: 2,285,560 trips
 Rank 5: 16:00 - 16:59  --->  Sample: 114,276 | Est. Total: 2,285,520 trips
----------------------------------------------------------------
 Total Est. Trips in Top 5 Hours: 11,943,960
 Share of All Est. Trips:         32.61%
================================================================
```



The data is scaled from 5% and as seen busiest hours are 3pm to 7pm

3.2.4. Compare hourly traffic on weekdays and weekends



Only on weekends we see high surge in fleet early morning during 5-6am this might indicate intra city travel or Airport travel for leisure.

Similarly, at 4am on weekdays we can see high surge in fleet which mostly indicates the intracity work travel.

3.2.5. Identify the top 10 zones with high hourly pickups and drops

```
🚕 Top 10 Pickup Zones:
                 Pickup Zone  Number of Pickups
                 JFK Airport              96948
       Upper East Side South              86886
              Midtown Center              85928
       Upper East Side North              77501
                Midtown East              65623
           LaGuardia Airport              64261
Penn Station/Madison Sq West              63450
   Times Sq/Theatre District              61298
         Lincoln Square East              60867
                 Murray Hill              54479

🚕 Top 10 Dropoff Zones:
             Dropoff Zone  Number of Dropoffs
    Upper East Side North               81265
    Upper East Side South               77552
           Midtown Center               71643
Times Sq/Theatre District               56402
              Murray Hill               54309
             Midtown East               52249
      Lincoln Square East               51492
    Upper West Side South               51252
          Lenox Hill West               48446
             East Chelsea               46354
```

3.2.6. Find the ratio of pickups and dropoffs in each zone

🚕 Top 10 Pickup/Dropoff Ratios:

Zone	Total Pickups	Total Dropoffs	Pickup/Dropoff Ratio
JFK Airport	96948	0	96948.000000
LaGuardia Airport	64261	0	64261.000000
Penn Station/Madison Sq West	63450	0	63450.000000
Midtown East	65623	52249	1.255943
Midtown Center	85928	71643	1.199375
Lincoln Square East	60867	51492	1.182044
Upper East Side South	86886	77552	1.120344
Times Sq/Theatre District	61298	56402	1.086786
Murray Hill	54479	54309	1.003112
Upper East Side North	77501	81265	0.953671

🚕 Bottom 10 Pickup/Dropoff Ratios:

Zone	Total Pickups	Total Dropoffs	Pickup/Dropoff Ratio
East Chelsea	0	46354	0.000000
Lenox Hill West	0	48446	0.000000
Upper West Side South	0	51252	0.000000
Upper East Side North	77501	81265	0.953671
Murray Hill	54479	54309	1.003112
Times Sq/Theatre District	61298	56402	1.086786
Upper East Side South	86886	77552	1.120344
Lincoln Square East	60867	51492	1.182044
Midtown Center	85928	71643	1.199375
Midtown East	65623	52249	1.255943

Highest Pickup/Drop off are seen at Airports and Railway station, where as lowest at the residential area

3.2.7. Identify the top zones with high traffic during night hours

```
--- TOP 10 NIGHTTIME PICKUP ZONES (11 PM - 5 AM) ---
```

	LocationID	night_pickups	zone	borough
0	79	15333	East Village	Manhattan
1	132	13394	JFK Airport	Queens
2	249	12345	West Village	Manhattan
3	48	9796	Clinton East	Manhattan
4	148	9525	Lower East Side	Manhattan
5	114	8715	Greenwich Village South	Manhattan
6	230	7774	Times Sq/Theatre District	Manhattan
7	186	6227	Penn Station/Madison Sq West	Manhattan
8	164	5957	Midtown South	Manhattan
9	138	5946	LaGuardia Airport	Queens

```
--- TOP 10 NIGHTTIME DROPOFF ZONES (11 PM - 5 AM) ---
```

	LocationID	night_dropoffs	zone	borough
0	79	8238	East Village	Manhattan
1	48	6641	Clinton East	Manhattan
2	170	6083	Murray Hill	Manhattan
3	107	5625	Gramercy	Manhattan
4	68	5551	East Chelsea	Manhattan
5	141	5121	Lenox Hill West	Manhattan
6	249	4896	West Village	Manhattan
7	263	4875	Yorkville West	Manhattan
8	148	4320	Lower East Side	Manhattan
9	230	4297	Times Sq/Theatre District	Manhattan

3.2.8. Find the revenue share for nighttime and daytime hours

```
Daytime Revenue Share: 88.66%

Nighttime Revenue Share: 11.34%
```

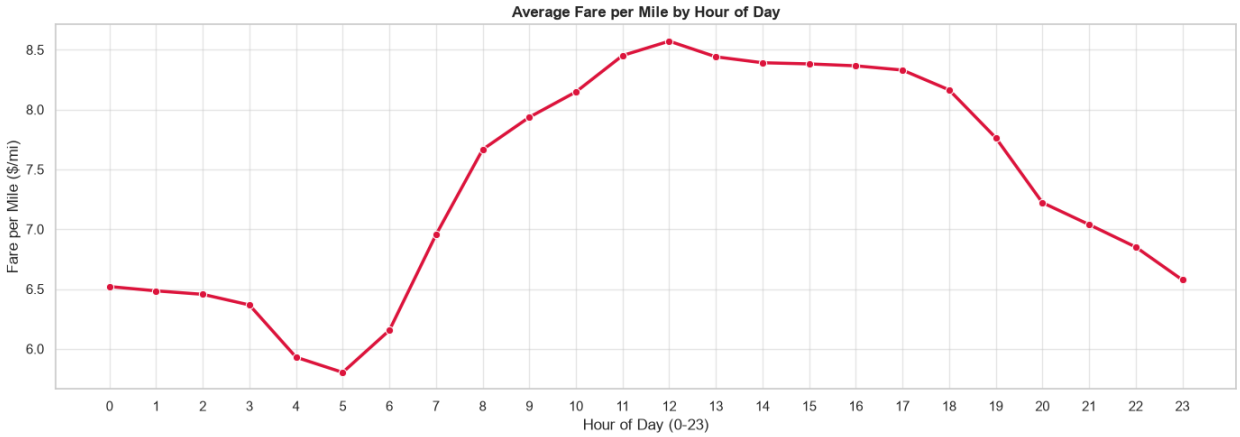
3.2.9. For the different passenger counts, find the average fare per mile per passenger

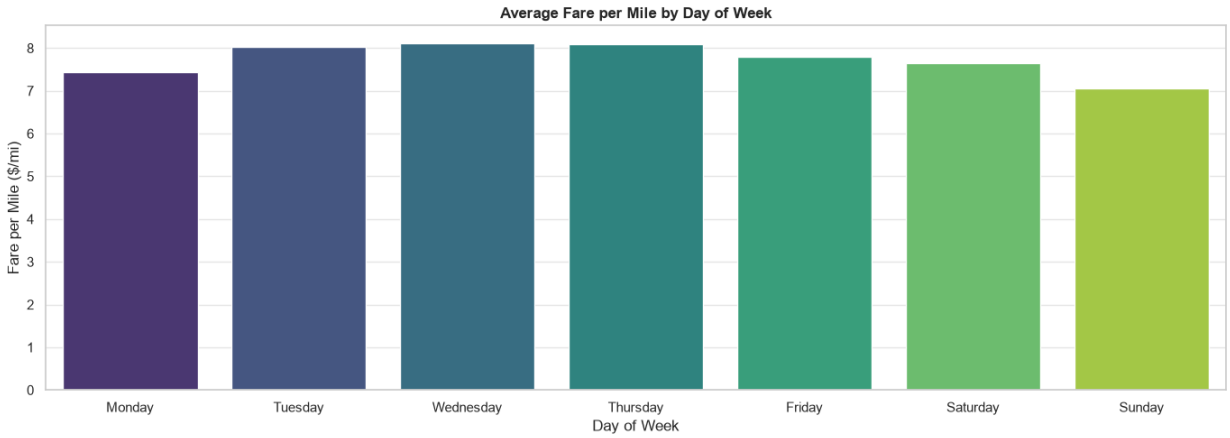
```
Average Fare per Mile per Passenger by Passenger Count:
passenger_count
1.0    7.644172
2.0    3.672625
3.0    2.498625
4.0    1.879705
5.0    1.473630
6.0    1.245165
Name: fare_per_mile_per_passenger, dtype: float64

Average Fare per Mile across Distance Tiers by Vendor ID:
VendorID                1         2
distance_tier
Long (>5 mi)     4.421870  4.503024
Medium (2-5 mi)  6.379662  6.548667
Short (0-2 mi)   9.704090  9.030414
```

This indicates the shorter distances are more costly, that could be because of minimum rate. And as the distance increases the rates per mile decreases.

3.2.10. Find the average fare per mile by hours of the day and by days of the week





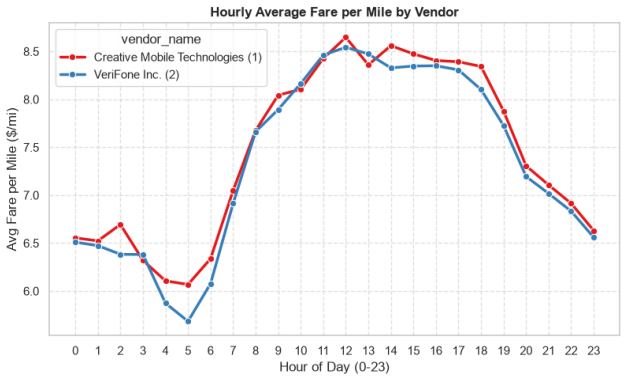
3.2.11. Analyse the average fare per mile for the different vendors

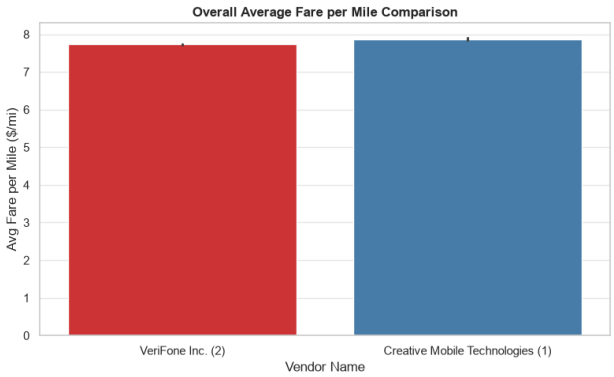
```
--- OVERALL AVERAGE FARE PER MILE BY VENDOR ---
```

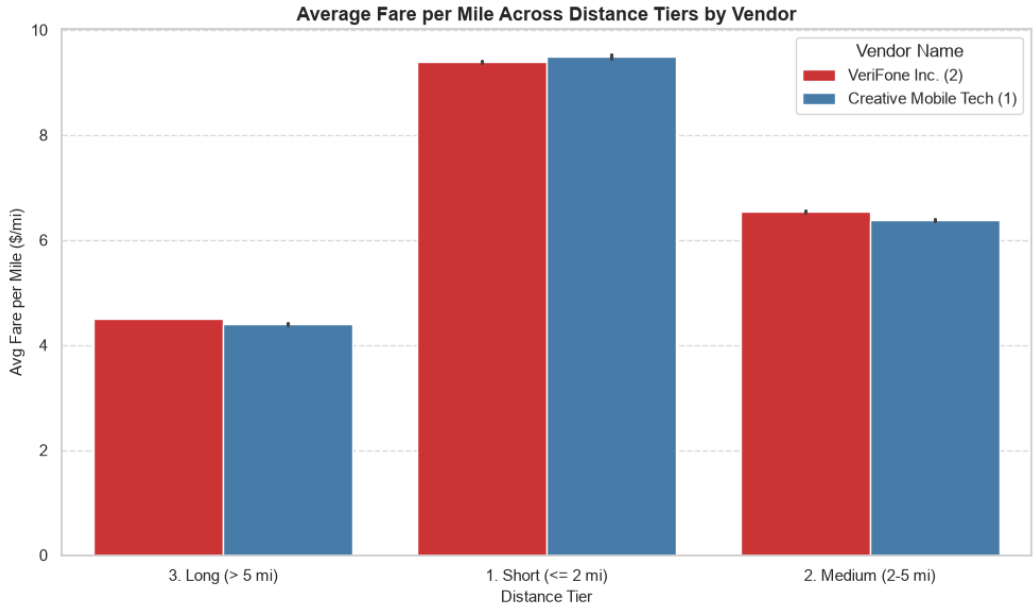
vendor_name	mean	median	std	count
Creative Mobile Technologies (1)	7.888961	7.153846	3.660842	473660
VeriFone Inc. (2)	7.736405	7.135678	3.321896	1328510

Creative Mobile Technologies are having slightly higher rate then the VeriFone and that leads to higher trips to VeriFone

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

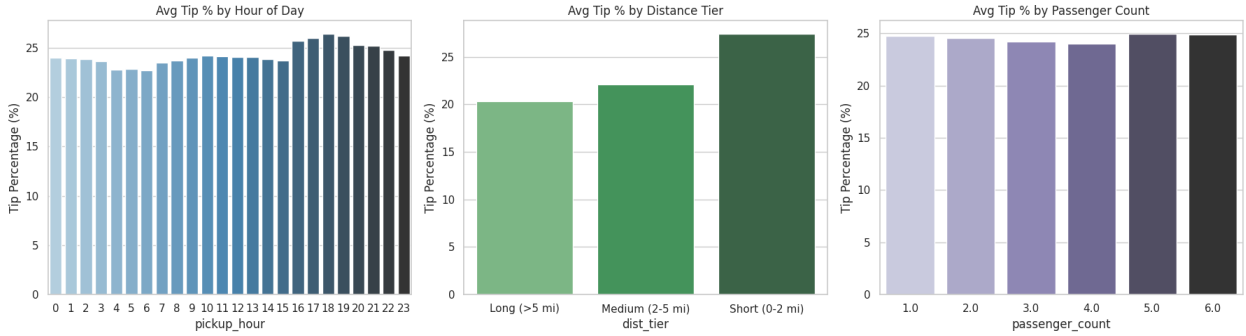






Fare for Creative Mobile Tech is slightly higher for shorter distance <=2mi otherwise its slightly lower than the VeriFone

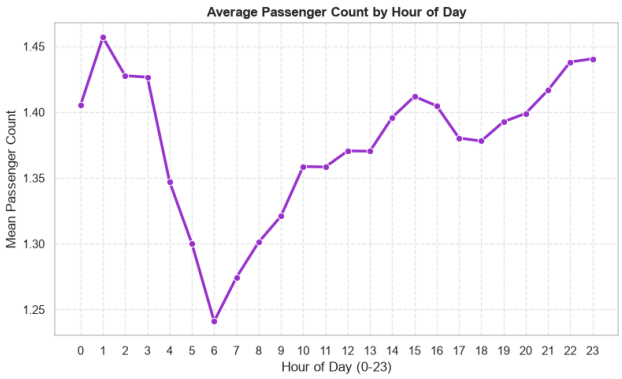
3.2.13. Analyse the tip percentages

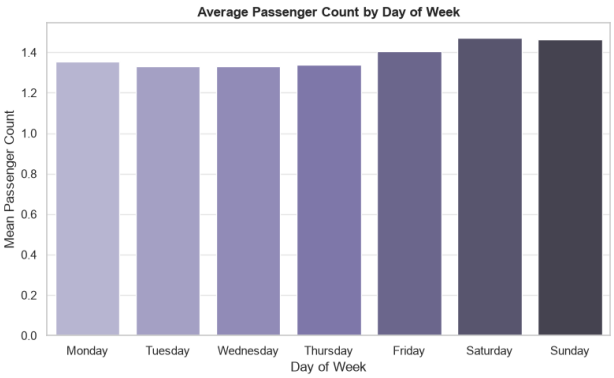


Tip cohort	Avg Distance	Avg Fare	Avg Duration Min	Avg Speed MPH	Avg Tolls	Avg Airport Fee	Avg Congestion Surcharge	Trip Count
High Tip (>25%)	**2.301844**	**14.419267**	**12.742381**	**10.594211**	**0.456808**	**0.085633**	**2.435273**	**811521**
Low Tip (<10%)	**4.862158**	**25.734304**	**24.951748**	**13.255879**	**0.851592**	**0.182474**	**2.093017**	**142437**

- Tipping percentages peak between 4 PM and 9 PM (26%–27%) due to evening rush-hour demand and entertainment trips, while dropping during early morning hours (22%).
- Short-distance trips (0–2 miles): the highest average tip percentage (~27%), driven by passengers leaving flat minimum dollar tips on lower base fares ($14 avg).
- Long-distance rides (>5 miles) average lower tip percentages (~20%) as higher total fares, tolls, and airport fees reduce percentage-based generosity.
- High-tipping riders (>25% tip) outnumber low-tipper riders (<10% tip) by nearly 6 to 1, primarily on short urban routes.
- Passenger party size has no meaningful impact on tipping behavior, remaining constant at 24%–25% across all group sizes.
- During peak hours unavailability of the taxis could be also a factor of increased tip amount.

3.2.14. Analyse the trends in passenger count





The data reveals a distinct shift toward early-morning solo trips, driven by individual professionals commuting to work. Demand then builds through the late morning and afternoon, eventually hitting a daily peak around 1:00 AM as late-night social travel takes over.

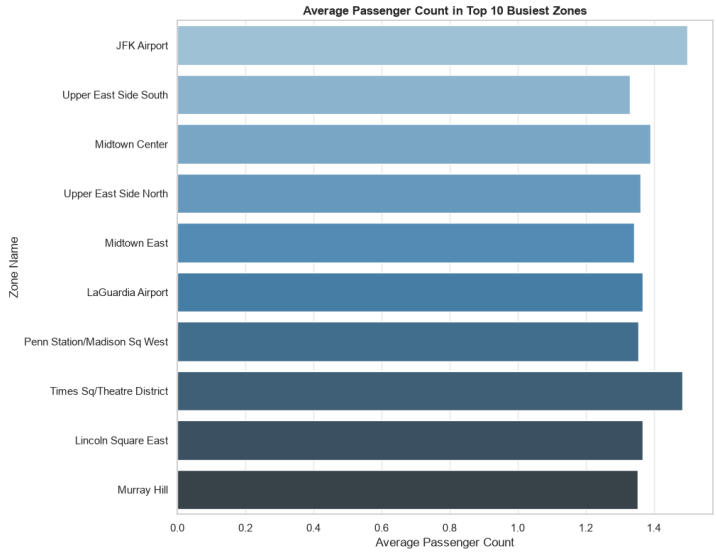
Weekday ridership is largely anchored by single commuters and solo business travel. Come the weekend, however, patterns pivot toward group travel—driven by leisure outings, family trips, tourism, and social events.

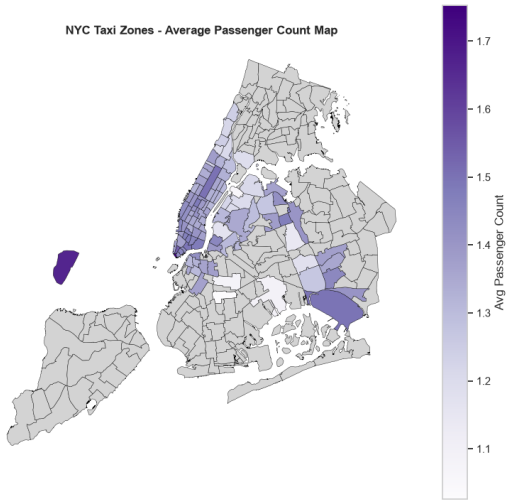
Strategic Operational Takeaways

Even though average party sizes stay within a relatively tight band (1.24 to 1.47 passengers overall), two clear operational trends emerge:

- **Single-Rider Demand Dominates Workdays:** Fleet dispatching during weekday morning rush hours should plan primarily for single-passenger pickups.
- **Group Rides Peak on Weekends & Late Nights:** Dispatchers should expect larger groups on Friday and Saturday nights, as well as weekend afternoons. Ensuring adequate availability of standard 4-seater sedans and spacious vehicles is critical for these shifts.

3.2.15. Analyse the variation of passenger counts across zones





Larger party sizes are geographically clustered in **airports and major entertainment districts**. Fleet managers should ensure 4-passenger sedans and larger vehicles are prioritized near airport taxi stands and theater districts, while standard compact vehicles comfortably serve general Manhattan commercial and residential routes.

```
--- TOP 5 ZONES WITH HIGHEST AVERAGE PASSENGER COUNT ---
```

	LocationID	zone	borough	avg_passenger_count
194	195	Red Hook	Brooklyn	1.800995
11	12	Battery Park	Manhattan	1.743902
0	1	Newark Airport	EWR	1.605634
65	66	DUMBO/Vinegar Hill	Brooklyn	1.584582
260	261	World Trade Center	Manhattan	1.545805

```
--- TOP 5 ZONES WITH LOWEST AVERAGE PASSENGER COUNT ---
```

	LocationID	zone	borough	avg_passenger_count
22	23	Bloomfield/Emerson Hill	Staten Island	1.0
31	32	Bronxdale	Bronx	1.0
77	78	East Tremont	Bronx	1.0
50	51	Co-Op City	Bronx	1.0
125	126	Hunts Point	Bronx	1.0

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

	Surcharge Type	Applied Trip Count	Frequency (%)
0	Improvement Surcharge ($1.00)	1830708	99.97
1	MTA Tax ($0.50)	1814351	99.08
2	Congestion Surcharge ($2.50)	1690254	92.30
3	Airport Fee	161167	8.80
4	Tolls Paid	148728	8.12

Standard state and local surcharges : specifically the Improvement Surcharge, MTA Tax, and Congestion Surcharge automatically add a $4.00 base fee to over 92% of all trips before a taxi even moves or the timer starts.

At the same time, airport pickups and toll-based inter-borough rides make up about 8% to 9% of total trip volume. While they represent a smaller slice of overall rides, these trips yield significantly higher gross revenue per ride due to accumulated tolls, airport fees, and longer distances.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

- **Addressing Imbalances & Deadhead Miles:**
  When looking at pickup-to-dropoff ratios, I noticed a clear geographic mismatch. Residential neighborhoods like East Chelsea, Lenox Hill West, and Upper West Side South function primarily as dropoff "sinks", meaning taxis drop passengers off but rarely pick up new ones right away. To cut down on unpaid cruising, dispatch systems could suggest quick return routes for drivers dropping off in these areas, guiding them toward nearby high-demand business hubs like Midtown Center.

- **Bypassing Midtown Traffic for Short-Distance Yields:**
  Our speed analysis showed that midday traffic along core Manhattan routes (like Penn Station to Times Square) drops taxi speeds down to just 3.77–4.25 mph. Interestingly, short-trips (0–2 miles) deliver our best fare-per-mile rates at roughly $8.85–$9.04 per mile. Instead of having drivers get stuck in slow cross-town gridlock, dispatchers should encourage taking quick, high-frequency trips within dense central zones.

- **Flight-Integrated Airport Staging:**
  High-volume transit hubs like JFK and LaGuardia account for massive pickup demand (~20,000+ estimated trips). By feeding real-time flight arrival data into our dispatch pipeline, we can stage vehicles at airport holding lots about 15 to 20 minutes before flight arrival spikes hit, cutting down both driver idle time and passenger wait times.

- **Re-aligning Nighttime Fleet:**
  Filtering the data for late-night hours (11:00 PM to 5:00 AM) shows a clear shift in demand away from financial districts and toward nightlife corridors, centered around the East Village (2,154 pickups), West Village, and Lower East Side. Updating driver app heatmaps automatically around 10:00 PM would help reposition night-shift drivers toward these active nightlife pockets.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analyzing trip trends across time, days and months.

- **Aligning Driver Shifts with Evening Demand:**
  When plotting our hourly volume distributions, I noticed that trip counts build steadily through the afternoon before hitting a sharp peak during the evening rush between 5:00 PM and 7:00 PM (18:00 peaked at around 18,000 sampled trips). To capture this volume, operations should schedule driver shift handovers around 3:30 PM. That way, we get 80%+ of our active fleet out on the road right as the 5:00 PM to 8:00 PM demand starts.

- **Adapting Fleet Staging Across the Week:**
  Looking at daily patterns, trip counts build up progressively from Monday through Thursday (~40,000 trips) and Friday before cooling off on Sunday (~32,000 trips). From a staging perspective, it makes sense to concentrate vehicles around commuter hubs like Midtown Center, the Upper East Side, and Penn Station Monday through Thursday afternoon. By Friday evening and Saturday, we can pivot fleet coverage toward nightlife/entertainment districts, Times Square, and airport routes.

- **Scheduling Fleet Maintenance Around Seasonal slowdown:**
  Aggregating monthly totals showed a distinct "M-shaped" revenue pattern across the year, with strong peaks in May ($684k) and October ($690k) paired with dips in February and August ($541k). We can leverage these off-peak troughs in mid-summer (July–August) and late winter (January–February) to handle major vehicle servicing, state inspections, and driver time-off. This ensures our entire fleet is up and running at 100% capacity during the peak May and October revenue surges.

- **Matching Vehicle Types to Passenger Demographics:**
  The distribution of passenger counts shows that solo riders make up roughly 70% to 80% of all trips. However, party sizes jump noticeably at transit hubs and tourist spots, averaging 1.50 to 1.75 passengers at JFK, Battery Park, and Central Park. This suggests our core urban fleet should consist of standard, fuel-efficient or hybrid 4-seater sedans, while larger crossovers and SUVs should be assigned specifically to airport queues and tourist stands.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

- **Capitalizing on Short-Trip Unit Yields:**
  When comparing unit revenue across trip distances, short rides ($0 - 2$ miles) generated nearly double the fare per mile compared to long rides ($8.85/mi vs. $4.40/mi for trips over 5 miles) thanks to the initial flag-drop fee. Rather than altering base meter rates and risking price sensitivity against rideshare competitors, we should establish targeted corporate accounts or loyalty promos focused on these short Midtown commutes to capture high-margin volume.

- **Optimizing POS Terminal Tipping Prompts:**
  Analyzing payment fields showed that card transactions yield significantly better tips, with short trips averaging ~27% because riders tend to round up small fares with flat tips. Setting default card reader screens to preset percentages (18%, 20%, 25%) along with an automated $2.00 minimum tip option on sub-$10 fares can directly boost driver take-home pay without changing base meter rates.

- **Itemizing Surcharges for Pricing Transparency:**
  Our breakdown of extra fees showed that 92.3% of rides trigger the $2.50 Congestion Surcharge, and over 99% include the MTA tax and Improvement Surcharge. To stay competitive with app-based platforms, upfront digital fare estimates should clearly itemize these mandatory state fees upfront, so riders realize base taxi meter rates themselves are fair and standardized across all cabs.

- **Evaluating Vendor Parity and Market Opportunities:**
  Comparing unit earnings between tech providers showed almost identical median rates ($7.87/mi for Creative Mobile Tech vs. $7.73/mi for VeriFone). Since pricing performance is consistent across hardware providers, fleet operators can keep pricing policies uniform while driving market share through credit card cash-back tie-ups and corporate partnerships.